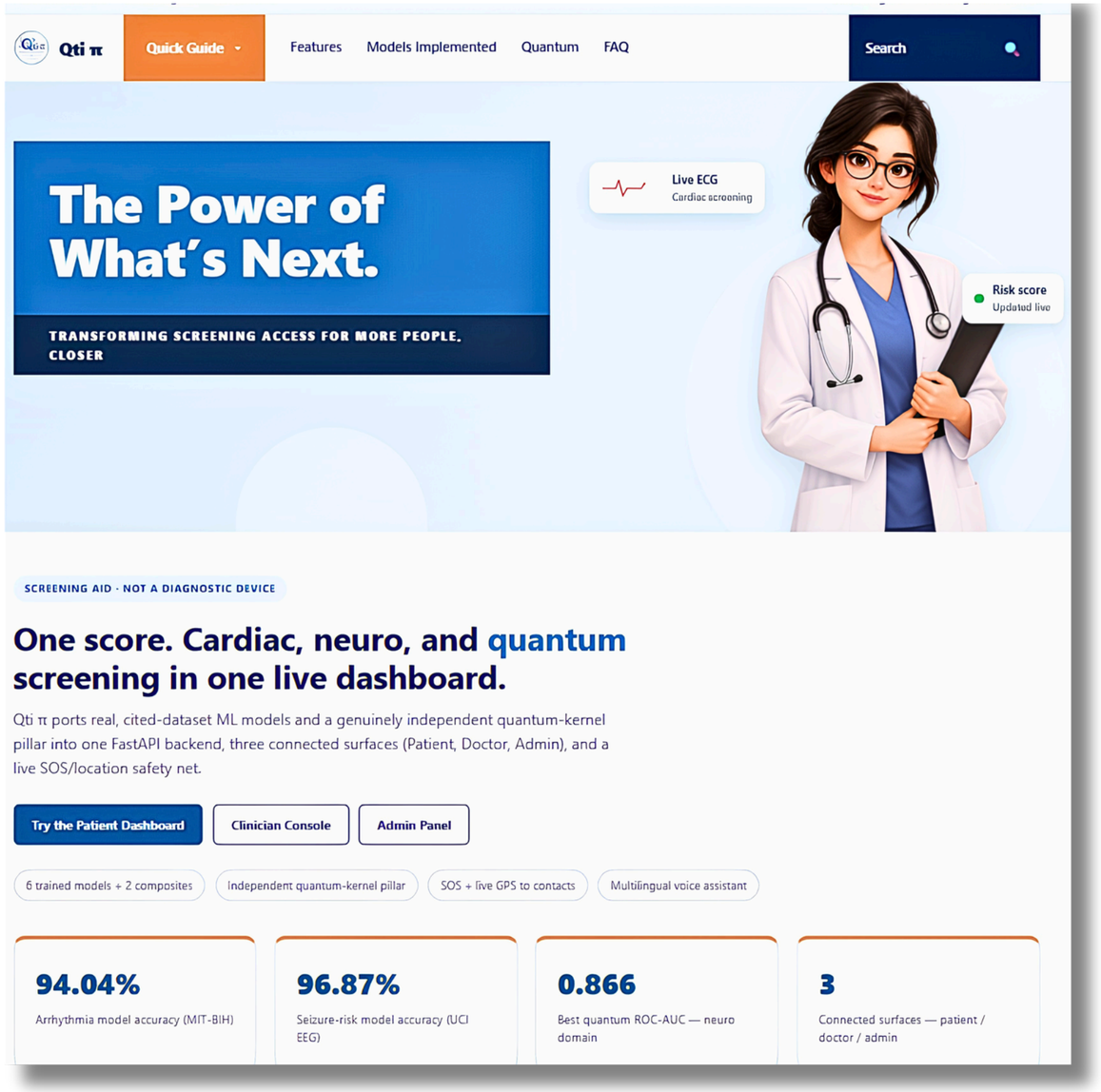
Team Name:



Qti π

- **Team Leader Name :** Mr. Ashmit Paul
- **Problem Statement Title-** Hybrid Quantum Machine Learning Platform for Early Disease Detection

PROPOSED SOLUTION

OUR SOLUTION TO IT...

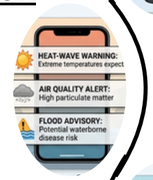
Qti Pi is a working platform today — not just a theoretical concept on paper



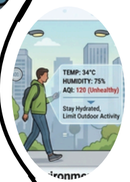
1.Continuos Health Monitoring



2. AI-Based Health Anomaly Detection



3. Disaster-Specific Health Alerts



4.Environmental Awareness



8. Scalable Deployment



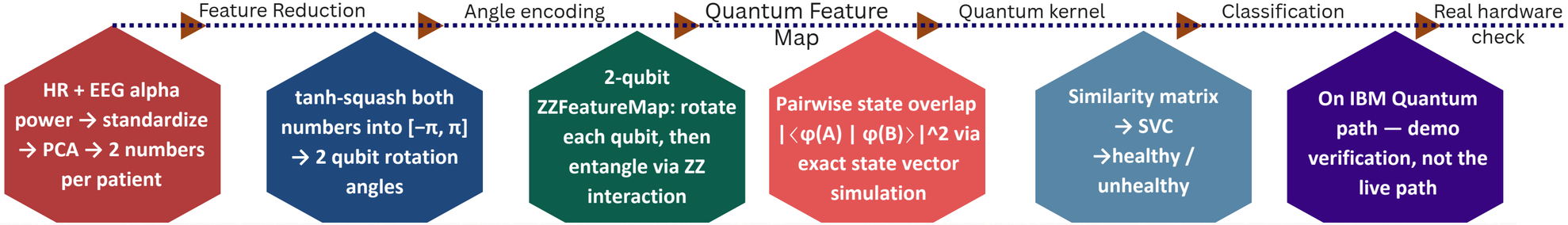
6. Emergency Assistance Features

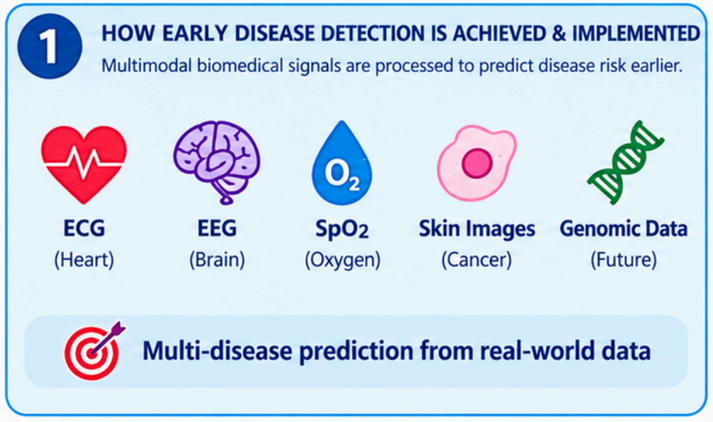


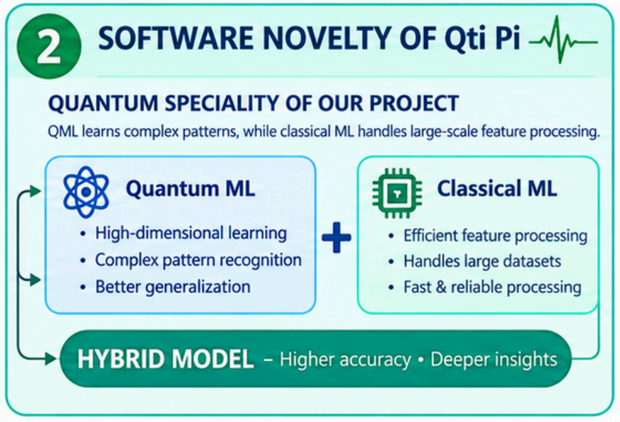
7. Privacy-Preserving Edge AI

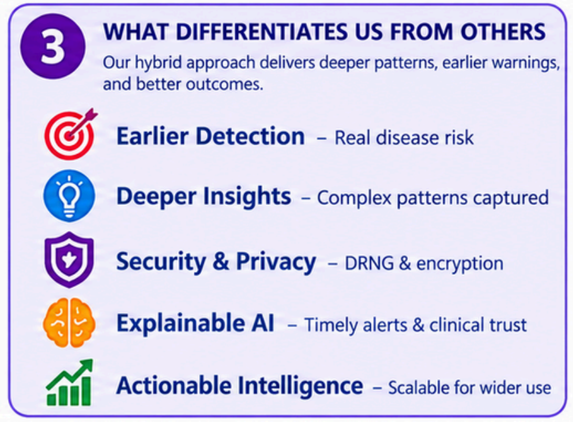


6 STEPS - QUANTUM ARCHITECHTURE

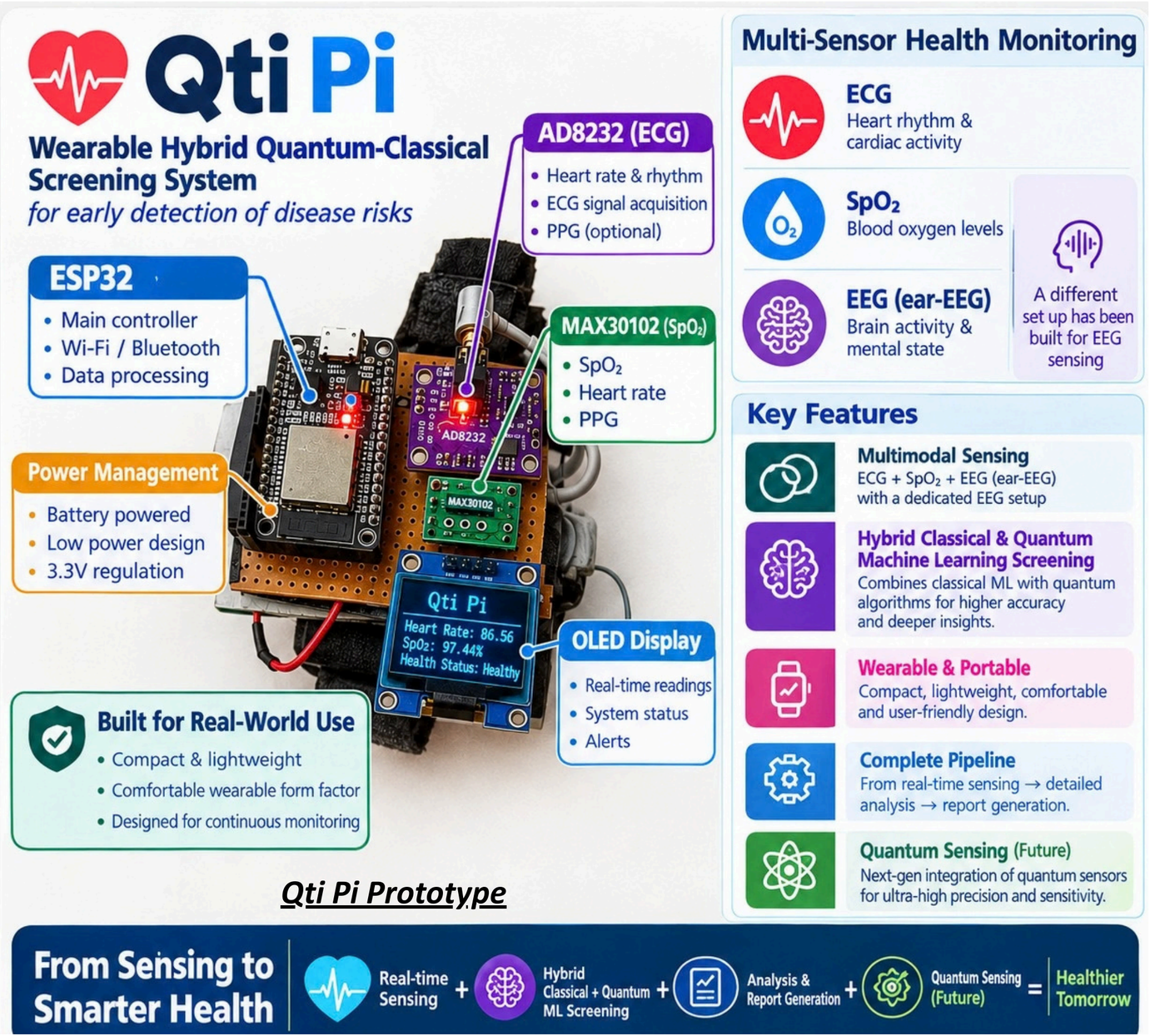




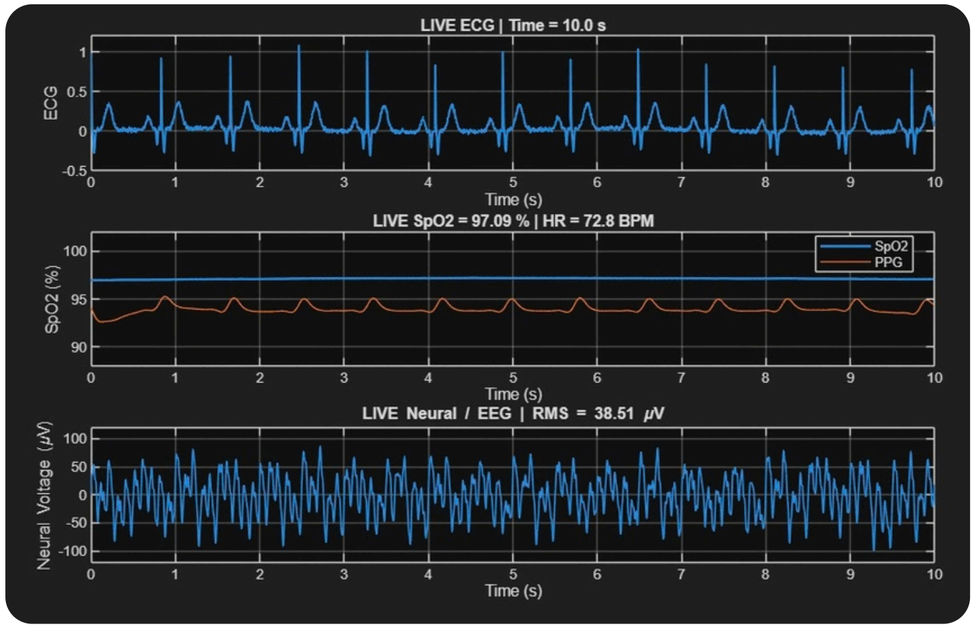




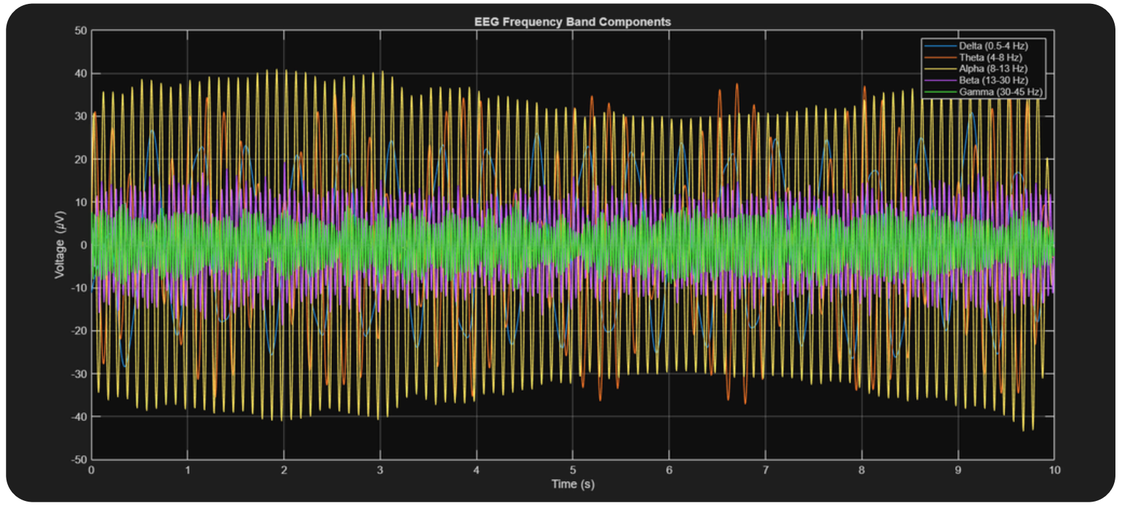
Qti Pi PROTOTYPE



Qti Pi Prototype



LIVE ECG,EEG &SPO2 HEALTH MONITORING



Brain Frequency Classification from brain voltage

TECHNICAL APPROACH

User wears Qti Pi wearable

ECG, EEG, SPo2 parameters sensed

All Sensing modules connected to ESP-32

ESP-32 interfaced with MATLAB

Bio-Signal Acquisition

Other health parameters calculated

Live Data fed into backend (FastAPI + ngrok)

Skin images uploaded directly on App

ML + QML Prediction

Detailed Analysis
Trend Analysis &
Classical v/s Quantum Disagreement
Score + Reasoning

Patient's Dashboard
Qti Pi Mobile App

Doctor's Dashboard

PDF
Detailed Report, generated & provided via Mail

Quantum Encryption

QRNG technology for secure key generation (AES 256)

TLS/FastAPI Backend

Pan–Tompkins

RR-Interval Analysis

Welch PSD

Band-Power Ratios

Ratio-of-Ratios

Band Pass filtering

Digital Lock-in Amplification

Median Denoising

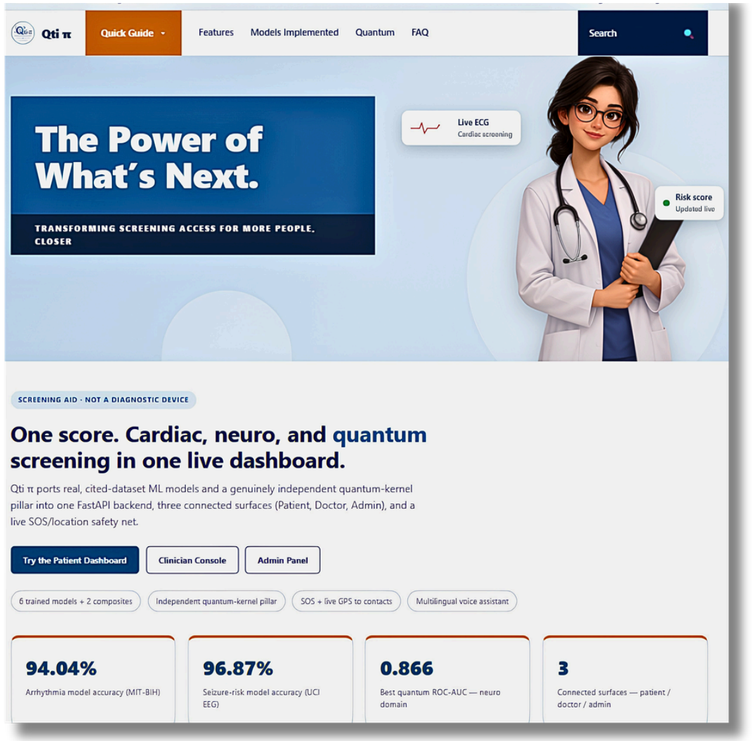
ML + QML Prediction

FEATURE PREPARATION
ECG + HR/HRV + SpO_2 + EEG Features

FEATURE REDUCTION
Normalization → PCA → Selected Features

Classical ML: Random Forest / HGB Risk Prediction
Quantum ML: Feature Encoding VQC / QSVM / QNN

Classical + Quantum Results
FINAL RISK SCORE



Qti π Patient's Dashboard
(Desktop View)

Tech Stack

CSS
JavaScript
HTML
MATLAB
ngrok

INNOVATION & DIFFERENTIATION

Condition	Model	Details	Dataset(s)
Heart disease	CardioRisk-GB	Accuracy 85.33% · Precision 87.88%	UCI Heart Failure Prediction (Cleveland + Hungary + Switzerland + VA Long Beach + Statlog merge)
Arrhythmia	RhythmNet-5	Accuracy 94.04% · Precision 94.74%	MIT-BIH Arrhythmia Database (PhysioNet)
Stroke risk	StrokeProxy-RF	Accuracy 69.78% · Recall 78.12%	CDC NHANES (multi-cycle, 2005–2020)
Seizure risk	NeuroScreen-RF	Accuracy 96.87% · Precision 97.09%	UCI Epileptic Seizure Recognition (Bonn University EEG database)
Heart failure	HF-Composite	Rule-weighted: CardioRisk-GB + RhythmNet-5 + BMI/obesity screen	UCI Heart Failure Prediction + MIT-BIH Arrhythmia Database
Neuro health	NeuroHealth-Composite	Rule-weighted: NeuroScreen-RF + tremor rule engine	UCI Epileptic Seizure Recognition (tremor part is rule-based)
Epilepsy	NeuroScreen-RF	Accuracy 96.87% · Precision 97.09%	UCI Epileptic Seizure Recognition
Skin lesion screening	DermaNet-QCNN	Accuracy 86.12% · Precision 87.36%	HAM10000
Diabetes screening	Diabetes-Tiered (1/2)	Tier 1 (vitals) 80.0% · Tier 2 (lab-assisted) 94.4–95.2%	CDC NHANES (2005–06 through 2017–Mar 2020)

Computational Efficiency Benchmarking

Achieved

Performance and efficiency metrics were successfully benchmarked.

Simulation Verification

Not Always Possible

Due to the 10 minute free cloud services constraint, all simulations could not be verified.

IBM Quantum Cloud Computing Services

10 minutes
Free cloud service limit

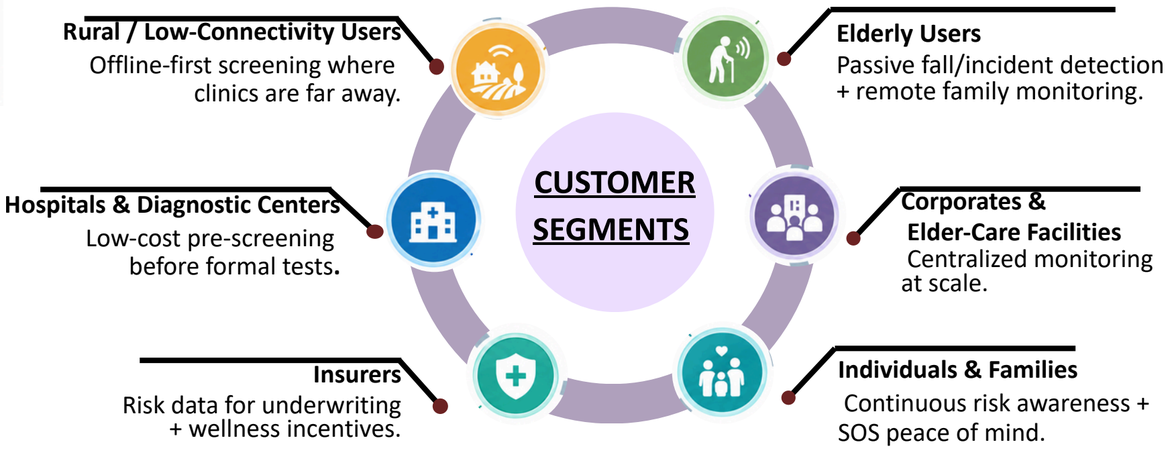
LINKS

- TAM SAM SOM Documentation
- Budget Report
- Survey Analysis

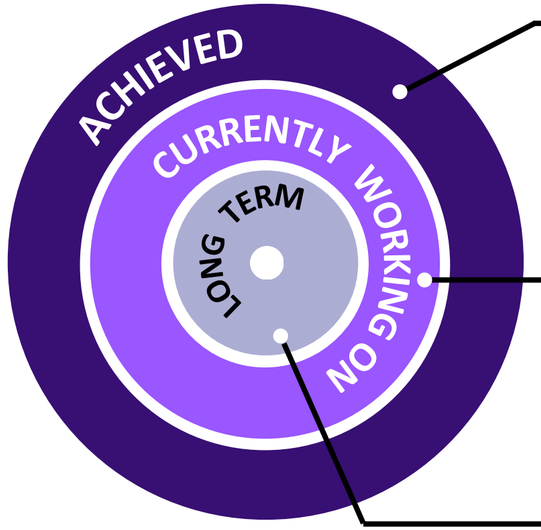
Competitive Analysis

Quick comparison of key capabilities, technology and focus

Capability	Our Project	Consumer Wearables	Single-Disease Apps	Quantum Prototypes
Data Inputs	HR, SpO2, ECG, pulse interval	2–4 signals, mostly HR/SpO2	1 signal, disease-specific	1–2 signals, lab-controlled
Disease Coverage	Cardiac + neuro (cancer roadmap)	None (fitness only)	One (arrhythmia or seizure)	One (proof-of-concept)
Quantum Layer	4-qubit VQC (IBM Quantum cloud)	None	None	Simulated only (not cloud)
Explainability	SHAP + classical/NN lean (quantum)	None	Minimal, black-box	Rarely reported
Safety Loop	AUAP SOS + contacts + voice assistant	Fall/irregular-rhythm alerts only	None, diagnostic only	None
Cloud Ready	Yes (offline-capable)	Yes	Varies	No (quantum access)
Maturity	End-to-end (quantum module active)	Shipping product	Shipping (narrow scope)	Academic / early research



FEASIBILITY, IMPACT & FUTURE



ACHIEVED
Cardiac & neuro monitoring, QNN-based skin-cancer detection, risk scoring, SOS & voice assistant. Classical ML: 94% accuracy; QSVM benchmarked and its low-AUC root cause documented. Quantum cryptography + QRNG added for data security.

CURRENT WORKING ON.....
Expand cardiac + EEG datasets, validate and benchmark Classical/Quantum ML, and complete PCB development. Train Classical & Quantum ML models on genomic data for improved disease prediction.

LONG TERM
Multi-disease screening platform: cardiac, neuro, cancer & diabetes, with 50+ cancer types via genomic datasets. Future: NV-centre & Rydberg quantum sensing for precision bio-signals detection.

Aligned with UN Sustainable Development Goals

SDG 3 – Good Health & Well-being
Target 3.4

Reduces premature mortality from non-communicable diseases through prevention and treatment — by continuous screening and escalation of high-risk cases.

SDG 10 – Reduced Inequalities
Target 10.2

Offline-first design serves elderly and rural users who live far from specialists, so core alerts work without internet access.

SDG 9 – Industry, Innovation & Infrastructure
Target 9.5

Advances technological capability through hybrid quantum-classical ML on real IBM Quantum hardware and post-quantum secured health data transfer (ML-KEM), on locally buildable hardware.



User Journey — From Reactive Symptom Response to Proactive Early Detection

Instead of waiting for symptoms, Qti Pi finds risks early — before you even feel sick.

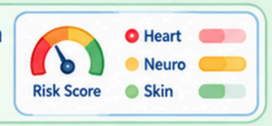
Before Qti Pi
(Reactive – We wait for symptoms)

1. **Patient feels fine day-to-day**
   No continuous health signal exists.
2. **Disease develops silently**
   (cardiac, neuro, or skin) with no early warning.
3. **Symptoms appear only**
   at a late, often acute stage.
4. **Patient reacts – visits hospital**
   after the event, not before it.
5. **Diagnosis and treatment**
   begin late → higher cost, worse outcome.

Qti Pi changes this

After Qti Pi
(Proactive – Finds risks early)

1. **Wearable continuously captures**
   ECG, PPG/SpO_2, and EEG (plus lesion photo upload).
2. **Real-time features extracted**
   on-device and streamed to the dashboard (ESP32 + Wi-Fi → FastAPI).
3. **Classical + Quantum ML**
   (QSVM/QNN/VQC) score risk across 7 conditions.
4. **Explainable risk score shown**
   (SHAP / ablation) → patient & clinician see WHY it fired.



5. **Threshold breach → automatic**
   alert + SOS escalation → early, proactive care.



Early Risk Detection · **Multi-Disease Screening** · **Affordable & Accessible Screening** · **Personalized Health Insights** · **Hybrid AI Reliability Signal** · **Wearable & Portable** · **Scalable Architecture** · **Future Quantum Sensing**

REFERENCES & RESEARCH

Quantum ML in Healthcare (motivation for the quantum module)

- Quantum ML for Colorectal Cancer Complication Prediction — arxiv.org/abs/2604.13951
- Quantum ML for Digital Health: A Systematic Review — arxiv.org/abs/2410.02446
- Evaluating QNN and QSVM Models in Healthcare — arxiv.org/abs/2505.20804

LINKS

- Demo Video
- Model Registry
- Backend ML/QML Metrics

Market Research (viability figures)

- Fortune Business Insights — Wearable Medical Devices Market Size, 2026
- Grand View Research — Wearable Medical Device Market Report, 2026-2033
- The Business Research Company — Wearable Heart Monitoring Devices Report, 2026
- Grand View Research (Wearable Cardiac Devices, 2025);
- IMARC Group (India Wearable Medical Devices, 2025);
- Grand View Research Horizon (India Patient Monitoring Devices, 2025)

Clinical Datasets (used to train existing models)

- MIT-BIH Arrhythmia Database — PhysioNet: https://physionet.org/content/mitdb/1.0.0/
- UCI Heart Failure Prediction Dataset — UCI Machine Learning Repository: https://archive.ics.uci.edu/dataset/45/heart+disease
- CDC NHANES (multi-cycle) — CDC/NCHS: https://www.cdc.gov/nchs/nhanes/index.html

Platforms & Policy

- IBM Quantum Cloud Platform — quantum.ibm.com
- Ayushman Bharat Digital Mission — abdm.gov.in
- National Quantum Mission, Dept. of Science & Technology, GoI — dst.gov.in
- Australian National University(ANU) QRNG Cloud Service — https://qrng.anu.edu.au/